

Proof Portfolio

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Theorem 1:

Let $f(x) = xe^{-x}$. Then for any positive integer x , the n -th derivative of f is given by:

$$f^{(n)} = (-1)^n e^{-x} (x - n)$$

Proof:

We proceed by induction on n , a positive integer.

Base Case: $n = 1$

We compute the first derivative

$$f'(x) = \frac{d}{dx}(xe^{-x}) = -xe^{-x} + e^{-x} = e^{-x}(1 - x)$$

Now compare with the right-hand side of the formula:

$$(-1)e^{-x}(x - 1) = e^{-x}(1 - x)$$

Thus, the formula holds for $n = 1$

Inductive Step:

Assume that the formula holds for some positive integer n . That is, that

$$f^{(n)}(x) = (-1)^n e^{-x} (x - n)$$

We want to show that the formula also holds for the $n + 1$ case. That is, that

$$f^{(n+1)}(x) = (-1)^{n+1} e^{-x} (x - n - 1)$$

Let us differentiate the inductive hypothesis:

$$f^{(n+1)}(x) = \frac{d}{dx}[(-1)^n e^{-x} (x - n)]$$

Applying the product rule gives:

$$f^{(n+1)}(x) = (-1)^n \left[\frac{d}{dx}(e^{-x})(x - n) + e^{-x} \frac{d}{dx}(x - n) \right]$$

This results in:

$$f^{(n+1)}(x) = (-1)^n [-e^{-x}(x-n) + e^{-x}] = (-1)^n e^{-x} [-(x-n)+1] = (-1)^n e^{-x} (-x+n+1)$$

Factoring out a negative one gives us:

$$f^{(n+1)}(x) = (-1)^{n+1} e^{-x} (x - n - 1)$$

Conclusion:

By the principle of mathematical induction, the formula:

$$f^{(n)}(x) = (-1)^n e^{-x} (x - n)$$

Holds for all positive integers n . ■

Theorem 2:

Let n be any integer greater than zero. Then, the following statement is true:

$$\sum_{k=1}^n \frac{1}{\sqrt{k}} \leq 2\sqrt{n}$$

Proof:

We proceed by induction on n , a positive integer.

Base Case: $n = 1$

We compute both sides of the inequality for this case:

$$\sum_{k=1}^1 \frac{1}{\sqrt{k}} = \frac{1}{1} = 1$$

$$2\sqrt{n} = 2\sqrt{1} = 2$$

$$1 \leq 2$$

Thus the statement is true for $n = 1$.

Induction Step:

Assume that the formula holds for some positive integer n . That is,

$$\sum_{k=1}^n \frac{1}{\sqrt{k}} \leq 2\sqrt{n}$$

We want to show that the formula also holds for the $n + 1$ case. That is,

$$\sum_{k=1}^{n+1} \frac{1}{\sqrt{k}} \leq 2\sqrt{n+1}$$

Let us substitute $n + 1$ into the left side of the induction hypothesis:

$$\sum_{k=1}^{n+1} \frac{1}{\sqrt{k}} = \sum_{k=1}^n \frac{1}{\sqrt{k}} + \frac{1}{\sqrt{n+1}}$$

After removing the last term, the sum matches the induction hypothesis. So we can see that

adding $\frac{1}{\sqrt{n+1}}$ to both sides of the induction hypothesis will give us:

$$\sum_{k=1}^n \frac{1}{\sqrt{k}} + \frac{1}{\sqrt{n+1}} \leq 2\sqrt{n} + \frac{1}{\sqrt{n+1}}$$

Which, by the prior statement is equivalent to:

$$\sum_{k=1}^{n+1} \frac{1}{\sqrt{k}} \leq 2\sqrt{n} + \frac{1}{\sqrt{n+1}}$$

Now, it becomes a matter of showing that the right side of this inequality is less than the right side of the inequality we wish to produce. That is, we want to show:

$$2\sqrt{n} + \frac{1}{\sqrt{n+1}} \leq 2\sqrt{n+1}$$

Let us consider this inequality to determine whether it holds true. Multiplying both sides by $\sqrt{n+1}$ gives:

$$2\sqrt{n(n+1)} + 1 \leq 2(n+1)$$

Simplifying and subtracting the one from both sides, we arrive at:

$$2\sqrt{n^2+n} \leq 2n+1$$

We then square both sides, resulting in:

$$4(n^2+n) \leq 4n^2+4n+1$$

$$4n^2+4n \leq 4n^2+4n+1$$

Finally, canceling like terms from both sides results in:

$$0 \leq 1$$

This statement is a *tautology*, and thus is always true for every value of n . Therefore, the following connection can be made by linking inequalities:

$$\sum_{k=1}^{n+1} \frac{1}{\sqrt{k}} \leq 2\sqrt{n} + \frac{1}{\sqrt{n+1}}$$

$$2\sqrt{n} + \frac{1}{\sqrt{n+1}} \leq 2\sqrt{n+1}$$

$$\sum_{k=1}^{n+1} \frac{1}{\sqrt{k}} \leq 2\sqrt{n} + \frac{1}{\sqrt{n+1}} \leq 2\sqrt{n+1}$$

Thus,

$$\sum_{k=1}^{n+1} \frac{1}{\sqrt{k}} \leq 2\sqrt{n+1}$$

Conclusion:

By the principle of mathematical induction, the following formula:

$$\sum_{k=1}^n \frac{1}{\sqrt{k}} \leq 2\sqrt{n}$$

Holds true for all positive integers n . ■

Theorem 3:

$$\binom{n^2}{2} = n \binom{n}{2} + n^2 \binom{n}{2}$$

Proof:

We shall prove this identity directly using algebraic simplification.

Left Hand Side:

$$\binom{n^2}{2} = \frac{(n^2)!}{2!(n^2-2)!} = \frac{(n^2)(n^2-1)}{2} = \frac{1}{2}n^4 - \frac{1}{2}n^2$$

Right Hand Side:

$$n \binom{n}{2} + n^2 \binom{n}{2}$$

Factoring and expanding:

$$(n + n^2) \frac{n!}{2!(n-2)!}$$

$$(n + n^2) \frac{1}{2} (n^2 - n) = (n^3 - n^2 + n^4 - n^3) \frac{1}{2} = \frac{1}{2}n^4 - \frac{1}{2}n^2$$

Conclusion:

So we have:

$$LHS = \frac{1}{2}n^4 - \frac{1}{2}n^2 = RHS$$

Therefore:

$$\binom{n^2}{2} = n \binom{n}{2} + n^2 \binom{n}{2} \quad \blacksquare$$

Theorem 4:

$$\frac{\binom{m \cdot n}{n} \binom{(m-1)n}{n} \binom{(m-2)n}{n} \dots \binom{n}{n}}{m!} = \frac{(m \cdot n)!}{(n!)^m \cdot m!}$$

Left Hand Side:

$$\frac{\binom{m \cdot n}{n} \binom{(m-1)n}{n} \binom{(m-2)n}{n} \dots \binom{n}{n}}{m!}$$

Expanding:

$$\frac{\frac{(m \cdot n)!}{n!(mn-n)!} \cdot \frac{((m-1) \cdot n)!}{n!((m-1)n-n)!} \cdot \frac{((m-2) \cdot n)!}{n!((m-2)n-n)!} \cdot \dots \cdot \frac{n!}{n!}}{m!}$$

Factoring:

$$\frac{\frac{(m \cdot n)!}{n!((m-1)n)!} \cdot \frac{((m-1) \cdot n)!}{n!((m-2)n)!} \cdot \frac{((m-2) \cdot n)!}{n!((m-3)n)!} \cdot \dots \cdot \frac{n!}{n!}}{m!}$$

Now, here we can see that each denominator matches the next fraction's numerator. We can cancel these in a sort of cascading fashion, ultimately cancelling $n!$ in the numerator of the last fraction when we finish. This cascading cancellation gives us the following simplified expression:

$$\frac{\frac{(m \cdot n)!}{(n!)^m}}{m!} = \frac{(m \cdot n)!}{(n!)^m \cdot m!}$$

Right Hand Side:

$$\frac{(m \cdot n)!}{(n!)^m \cdot m!}$$

Conclusion:

So we have:

$$LHS = \frac{(m \cdot n)!}{(n!)^m \cdot m!} = RHS$$

Therefore:

$$\frac{\binom{m \cdot n}{n} \binom{(m-1)n}{n} \binom{(m-2)n}{n} \dots \binom{n}{n}}{m!} = \frac{(m \cdot n)!}{(n!)^m \cdot m!} \quad \blacksquare$$

Theorem 5:

Any simple connected planar graph must have a vertex of degree 5 or less.

Proof:

Let G be any simple connected planar graph. By contradiction, let us assume that every vertex in G has degree 5 or greater. Thus, the sum of degrees must be constrained by:

$$\sum_{v \in V} \deg(v) \geq \sum_{v \in V} 5 \geq 5|V|$$

Now recall the 1st Handshaking Lemma:

$$\sum_{v \in V} \deg(v) = 2|E|$$

Thus, combining these, we can see:

$$2|E| \geq 5|V|$$

$$|E| \geq \frac{5}{2}|V|$$

Where V is the vertex set of G , and E is the edge set of G .

Now, since G is a planar graph, the minimum degree of a given region is 3, and therefore the sum of degrees of regions can be described as:

$$\sum_{r \in F} \deg(r) \geq 3|F|$$

Where F is the face set of G .

Now recall the 2nd Handshaking Lemma:

$$\sum_{r \in F} \deg(r) = 2|E|$$

Therefore, the following substitution can be made:

$$2|E| \geq 3|F|$$

Now, recall Euler's formula:

$$|V| - |E| + |F| = 2$$

Solving for $|F|$:

$$|F| = 2 - |V| + |E|$$

Substituting this into the inequality:

$$2|E| \geq 3(2 - |V| + |E|)$$

$$2|E| \geq 6 - 3|V| + 3|E|$$

$$|E| \leq 3|V| - 6$$

So, by Planarity, we also have:

$$|E| \leq 3|V| - 6$$

Therefore,

$$3|V| \leq |E| \leq 3|V| - 6$$

This is a **contradiction** since $|E|$ cannot be both greater than $3|V|$ and less than $3|V| - 6$.

Conclusion:

Since the assumption that every vertex in a simple connected planar graph has a degree 5 or greater, this must be false. Therefore, any simple connected planar graph must have a vertex of degree 5 or less. ■

Theorem 6:

A simple graph with 17 vertices and 73 edges cannot be bipartite.

Proof:

Let G be a simple graph with 17 vertices and 73 edges. By contradiction, let us assume that G is bipartite. Then, let us divide the vertices into 2 sets, V_1 and V_2 , each having different colors.

Thus:

$$|V_1| + |V_2| = 17$$

Since G is bipartite and the vertices are separated by type, the maximum number of edges in the graph is $|V_1| \cdot |V_2|$ when all vertices of V_1 connect to all vertices of V_2 .

To find this maximum number of edges, we want to find the maximum of the function:

$$f(x, y) = x \cdot y$$

Subject to the constraint:

$$x + y = 17$$

To do this, we can use the method of Lagrange multipliers:

$$\nabla f = \lambda \nabla g$$

$$\langle y, x \rangle = \lambda \langle 1, 1 \rangle$$

$$x = \lambda, y = \lambda$$

$$x = y$$

Substitute into constraint:

$$x + x = 17$$

$$x = y = \frac{17}{2} = 8.5$$

Now, under the context of Calculus, we can have decimal values, but moving back to the context of discrete math and the sizes of sets, we cannot have a decimal value. So, by the discreteness of the sets, instead of having two sets each with 8.5 vertices (which doesn't make sense), we will have to have one of the sets, say V_1 with 9 vertices, and the other, say V_2 with 8 vertices. Having determined this, we now can calculate the maximum number of edges as:

$$|V_1| \cdot |V_2| = 9 \cdot 8 = 72$$

However, here we have reached a **contradiction**—the maximum number of edges in the graph, under the assumption that it is bipartite, is 72, which is smaller than the actual number of edges, 73.

Conclusion:

Since the assumption that G is bipartite led to this contradiction, this must be false, and therefore a simple graph with 17 vertices and 73 edges cannot be bipartite. ■